

**SAFETY DATA SHEET**

Creation Date 02-November-2009

Revision Date 15-December-2022

Revision Number 7

**1. Identification**

|                             |                                                |
|-----------------------------|------------------------------------------------|
| <b>Product Name</b>         | <b>Formic acid</b>                             |
| <b>CAS-No</b>               | 64-18-6                                        |
| <b>Synonyms</b>             | Methanoic acid                                 |
| <b>Recommended Use</b>      | Laboratory chemicals.                          |
| <b>Uses advised against</b> | Food, drug, pesticide or biocidal product use. |

**Details of the supplier of the safety data sheet****Company**

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

**Manufacturer**

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number**

CHEMTREC®, Inside the USA: 800-424-9300  
CHEMTREC®, Outside the USA: 001-703-527-3887

**2. Hazard(s) identification****Classification**

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                |              |
|------------------------------------------------|--------------|
| <b>Flammable liquids</b>                       | Category 3   |
| <b>Acute oral toxicity</b>                     | Category 4   |
| <b>Acute Inhalation Toxicity</b>               | Category 3   |
| <b>Skin Corrosion/Irritation</b>               | Category 1 A |
| <b>Serious Eye Damage/Eye Irritation</b>       | Category 1   |
| <b>Health Hazards Not Otherwise Classified</b> | Category 1   |
| <b>Corrosive to the respiratory tract</b>      |              |

**Label Elements****Signal Word**

Danger

**Hazard Statements**

Flammable liquid and vapor  
Harmful if swallowed

Toxic if inhaled  
Causes severe skin burns and eye damage  
Corrosive to the respiratory tract



### Precautionary Statements

#### Prevention

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting/equipment  
Use only non-sparking tools  
Take precautionary measures against static discharges  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection  
Wear respiratory protection

#### Response

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Rinse mouth  
Do NOT induce vomiting  
Wash contaminated clothing before reuse  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

#### Storage

Store in a well-ventilated place. Keep container tightly closed  
Store locked up

#### Disposal

Dispose of contents/container to an approved waste disposal plant

## 3. Composition/Information on Ingredients

| Component   | CAS-No  | Weight % |
|-------------|---------|----------|
| Formic acid | 64-18-6 | >95      |

## 4. First-aid measures

#### General Advice

Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

#### Eye Contact

In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | attention is required.                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Inhalation</b>                      | Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required. If not breathing, give artificial respiration.                                                                                |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                                                                |
| <b>Most important symptoms/effects</b> | Difficulty in breathing. Causes burns by all exposure routes. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                                                                                                         |

## 5. Fire-fighting measures

|                                         |                                                                                                                                         |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                |
| <b>Flash Point</b>                      | 50 °C / 122 °F                                                                                                                          |
| <b>Method -</b>                         | No information available                                                                                                                |
| <b>Autoignition Temperature</b>         | 520 °C / 968 °F                                                                                                                         |
| <b>Explosion Limits</b>                 |                                                                                                                                         |
| <b>Upper</b>                            | 45 vol %                                                                                                                                |
| <b>Lower</b>                            | 10 vol %                                                                                                                                |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                |

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen. Thermal decomposition can lead to release of irritating gases and vapors.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 3             | 2                   | 1                  | N/A                     |

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                                      |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges. |
| <b>Environmental Precautions</b>            | Should not be released into the environment. Do not flush into surface water or sanitary sewer system. See Section 12 for additional Ecological Information.                                                                                         |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.                                                                        |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Keep away from heat, sparks and flame. Containers should be vented periodically in order to overcome pressure buildup. Incompatible Materials. Strong oxidizing agents. Metals. Finely powdered metals. Strong bases.                                                                                   |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component   | Alberta                                                                                | British Columbia           | Ontario TWA/EV             | Quebec                                                                                 | ACGIH TLV                  | OSHA PEL                                                                                             | NIOSH                                                  |
|-------------|----------------------------------------------------------------------------------------|----------------------------|----------------------------|----------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Formic acid | TWA: 5 ppm<br>TWA: 9.4 mg/m <sup>3</sup><br>STEL: 10 ppm<br>STEL: 19 mg/m <sup>3</sup> | TWA: 5 ppm<br>STEL: 10 ppm | TWA: 5 ppm<br>STEL: 10 ppm | TWA: 5 ppm<br>TWA: 9.4 mg/m <sup>3</sup><br>STEL: 10 ppm<br>STEL: 19 mg/m <sup>3</sup> | TWA: 5 ppm<br>STEL: 10 ppm | (Vacated) TWA: 5 ppm<br>(Vacated) TWA: 9 mg/m <sup>3</sup><br>TWA: 5 ppm<br>TWA: 9 mg/m <sup>3</sup> | IDLH: 30 ppm<br>TWA: 5 ppm<br>TWA: 9 mg/m <sup>3</sup> |

#### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Face protection shield or Goggles

#### Hand Protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments                                         |
|----------------|-------------------|-----------------|--------------------------------------------------------|
| Neoprene       | > 480 minutes     | 0.5 mm          | As tested under EN374-3                                |
| Butyl rubber   | > 480 minutes     | 0.7 mm          | Determination of Resistance to Permeation by Chemicals |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143 Acid gases filter Type E Yellow conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

#### **Environmental exposure controls**

Prevent product from entering drains.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

### 9. Physical and chemical properties

|                                               |                              |
|-----------------------------------------------|------------------------------|
| <b>Physical State</b>                         | Liquid                       |
| <b>Appearance</b>                             | Colorless                    |
| <b>Odor</b>                                   | pungent                      |
| <b>Odor Threshold</b>                         | No information available     |
| <b>pH</b>                                     | 2.1 10 g/L aq.sol            |
| <b>Melting Point/Range</b>                    | 8 °C / 46.4 °F               |
| <b>Boiling Point/Range</b>                    | 101 °C / 213.8 °F @ 760 mmHg |
| <b>Flash Point</b>                            | 50 °C / 122 °F               |
| <b>Evaporation Rate</b>                       | No information available     |
| <b>Flammability (solid,gas)</b>               | Not applicable               |
| <b>Flammability or explosive limits</b>       |                              |
| <b>Upper</b>                                  | 45 vol %                     |
| <b>Lower</b>                                  | 10 vol %                     |
| <b>Vapor Pressure</b>                         | 44 mbar @ 20 °C              |
| <b>Vapor Density</b>                          | No information available     |
| <b>Specific Gravity</b>                       | 1.220                        |
| <b>Solubility</b>                             | miscible                     |
| <b>Partition coefficient; n-octanol/water</b> | No data available            |
| <b>Autoignition Temperature</b>               | 520 °C / 968 °F              |
| <b>Decomposition Temperature</b>              | No information available     |
| <b>Viscosity</b>                              | 1.47 mPa.s @ 20 °C           |
| <b>Molecular Formula</b>                      | C H2 O2                      |
| <b>Molecular Weight</b>                       | 46.02                        |

### 10. Stability and reactivity

|                                         |                                                                                                                                             |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                                                                                  |
| <b>Stability</b>                        | Hygroscopic. heat sensitive. Decomposes to water and carbon dioxide.                                                                        |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. Exposure to moist air or water.       |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Metals, Finely powdered metals, Strong bases                                                                       |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Hydrogen, Thermal decomposition can lead to release of irritating gases and vapors |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                                    |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                                               |

### 11. Toxicological information

#### **Acute Toxicity**

#### **Product Information**

**Oral LD50**

Category 4.

**Dermal LD50**

Based on ATE data, the classification criteria are not met.

## Vapor LC50

Category 3.

## Component Information

| Component   | LD50 Oral         | LD50 Dermal | LC50 Inhalation                    |
|-------------|-------------------|-------------|------------------------------------|
| Formic acid | 730 mg/kg ( Rat ) | Not listed  | 15 g/m <sup>3</sup> ( Rat ) 15 min |

## Toxicologically Synergistic Products

No information available

Delayed and immediate effects as well as chronic effects from short and long-term exposure

## Irritation

Causes severe burns by all exposure routes Irritating to respiratory system

## Sensitization

No information available

## Carcinogenicity

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component   | CAS-No  | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------|---------|------------|------------|------------|------------|------------|
| Formic acid | 64-18-6 | Not listed | Not listed | Not listed | Not listed | Not listed |

## Mutagenic Effects

No information available

## Reproductive Effects

No information available.

## Developmental Effects

No information available.

## Teratogenicity

No information available.

## STOT - single exposure

None known

## STOT - repeated exposure

None known

## Aspiration hazard

No information available

## Symptoms / effects, both acute and delayed

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

## Endocrine Disruptor Information

No information available

| Component   | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|-------------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| Formic acid | Applicable                               | Not applicable                                   | Not applicable                          |

## Other Adverse Effects

The toxicological properties have not been fully investigated.

## 12. Ecological information

## Ecotoxicity

Contains a substance which is: Harmful to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component   | Freshwater Algae   | Freshwater Fish                        | Microtox             | Water Flea         |
|-------------|--------------------|----------------------------------------|----------------------|--------------------|
| Formic acid | EC50 = 25 mg/L/96h | Leuciscus idus: LC50 = 46-100 mg/L/96h | EC50 = 46.7 mg/L/17h | EC50 = 34 mg/L/48h |

## Persistence and Degradability

Miscible with water Persistence is unlikely based on information available.

## Bioaccumulation/ Accumulation

No information available.

## Mobility

. Will likely be mobile in the environment due to its water solubility.

| Component   | log Pow |
|-------------|---------|
| Formic acid | -0.54   |

## 13. Disposal considerations

## Waste Disposal Methods

Chemical waste generators must determine whether a discarded chemical is classified as a

hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component             | RCRA - U Series Wastes | RCRA - P Series Wastes |
|-----------------------|------------------------|------------------------|
| Formic acid - 64-18-6 | U123                   | -                      |

## 14. Transport information

### DOT

|                         |             |
|-------------------------|-------------|
| UN-No                   | UN1779      |
| Proper Shipping Name    | FORMIC ACID |
| Hazard Class            | 8           |
| Subsidiary Hazard Class | 3           |
| Packing Group           | II          |

### TDG

|                         |             |
|-------------------------|-------------|
| UN-No                   | UN1779      |
| Proper Shipping Name    | FORMIC ACID |
| Hazard Class            | 8           |
| Subsidiary Hazard Class | 3           |
| Packing Group           | II          |

### IATA

|                         |             |
|-------------------------|-------------|
| UN-No                   | UN1779      |
| Proper Shipping Name    | FORMIC ACID |
| Hazard Class            | 8           |
| Subsidiary Hazard Class | 3           |
| Packing Group           | II          |

### IMDG/IMO

|                         |             |
|-------------------------|-------------|
| UN-No                   | UN1779      |
| Proper Shipping Name    | FORMIC ACID |
| Hazard Class            | 8           |
| Subsidiary Hazard Class | 3           |
| Packing Group           | II          |

## 15. Regulatory information

### International Inventories

| Component   | CAS-No  | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-------------|---------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Formic acid | 64-18-6 | X   | -    | X    | ACTIVE                                        | 200-579-1 | -      | -   |

| Component   | CAS-No  | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-------------|---------|-------|------|------|------|------|------|-------|-------|
| Formic acid | 64-18-6 | X     | X    | X    | X    | X    | X    | X     | X     |

### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component | Canada - National Pollutant | Canadian Environmental | Canada's Chemicals Management |
|-----------|-----------------------------|------------------------|-------------------------------|
|-----------|-----------------------------|------------------------|-------------------------------|

|             | Release Inventory (NPRI)                      | Protection Agency (CEPA)<br>- List of Toxic Substances | Plan (CEPA) |
|-------------|-----------------------------------------------|--------------------------------------------------------|-------------|
| Formic acid | Part 1, Group A Substance Part 4<br>Substance |                                                        |             |

## Other International Regulations

## Authorisation/Restrictions according to EU REACH

| Component   | REACH (1907/2006) - Annex XIV -<br>Substances Subject to<br>Authorization | REACH (1907/2006) - Annex XVII -<br>Restrictions on Certain Dangerous<br>Substances | REACH Regulation (EC<br>1907/2006) article 59 - Candidate<br>List of Substances of Very High<br>Concern (SVHC) |
|-------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Formic acid | -                                                                         | Use restricted. See item 75.<br>(see link for restriction details)                  | -                                                                                                              |

<https://echa.europa.eu/substances-restricted-under-reach>

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component   | CAS-No  | OECD HPV | Persistent Organic<br>Pollutant | Ozone Depletion<br>Potential | Restriction of<br>Hazardous<br>Substances (RoHS) |
|-------------|---------|----------|---------------------------------|------------------------------|--------------------------------------------------|
| Formic acid | 64-18-6 | Listed   | Not applicable                  | Not applicable               | Not applicable                                   |

| Component   | CAS-No  | Seveso III Directive<br>(2012/18/EC) -<br>Qualifying Quantities<br>for Major Accident<br>Notification | Seveso III Directive<br>(2012/18/EC) -<br>Qualifying Quantities<br>for Safety Report<br>Requirements | Rotterdam<br>Convention (PIC) | Basel Convention<br>(Hazardous Waste) |
|-------------|---------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------|
| Formic acid | 64-18-6 | Not applicable                                                                                        | Not applicable                                                                                       | Not applicable                | Annex I - Y34                         |

## 16. Other information

## Prepared By

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## Creation Date

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## Revision Summary

This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**